Investigation: PETRÓLEO BRASILEIRO S.A. - PETROBRAS (CVM code 9512), fiscal year 2010, as it was on 2011-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Petróleo e Gás (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2010 DFP, version 3, filed on 2011-03-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-02-25); 2 (2011-03-04); 3 (2011-03-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=5169

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.944 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 79 (n = 317)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.057 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 67 (n = 406)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.120 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 84 (n = 414)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.795 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 95 (n = 416)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.166 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value

Percentile in group and year: 56 (n = 414)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.197 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 79 (n = 327)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.967 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 50 (n = 411)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.034 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 48 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.755 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 8 (n = 424)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −11.6 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 16.9 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 20 (n = 391)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −6.79 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 6 (n = 383)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.040 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 45 (n = 428)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 3.406 (index)
Components: X1 = 0.096; X2 = 0.195; X3 = 0.088; X4 = 1.479 (indices)
Percentile in sector and year: 70 (n = 533); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA: —
- Cash above earnings (ACCRUAL): 1
- Change in leverage: —
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- FCA 2010, v1: 2010-05-31
- FCA 2010, v2: 2010-08-02
- DFP 2010, v1: 2011-02-25
- DFP 2010, v2: 2011-03-04
- DFP 2010, v3: 2011-03-04

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 31 material

fact documents and 188 market notice documents.
Material facts filed in the 24 months before the date (from 2009-05-01 to 2011-04-30):
- 2011-04-29: Distribuição de Juros Sobre Capital Próprio
- 2011-03-28: Pagamento de Juros sobre Capital Próprio
- 2011-03-01: Incorporação da Companhia Mexilhão do Brasil
- 2011-02-25: Distribuição de Dividendos e Juros Sobre Capital Próprio.
- 2011-01-14: Reservas Provadas da Petrobras em 2010.
- 2010-12-29: subject omitted (possible person's name); see the document at the CVM
- 2010-12-27: Pagamento de Juros sobre Capital Próprio.
- 2010-12-14: Incorporação da Comperj Petroquímicos Básicos S.A. e da Comperj PET S.A.
- 2010-12-14: Fato Relevante - Distribuição de Juros Sobre Capital Próprio
- 2010-12-14: subject omitted (possible person's name); see the document at the CVM
- 2010-11-23: Fato Relevante - Pagamento de Juros sobre Capital Próprio
- 2010-11-09: Incorporação da Marlim Participações S.A. e da Nova Marlim Participações S.A.
- 2010-10-29: Esclarecimento sobre volume de descoberta em Sergipe
- 2010-10-22: Distribuição de Juros Sobre Capital Próprio.
- 2010-09-23: Fato Relevante - Aprovação do Preço por Ação
- 2010-09-20: Esclarecimento Oferta Pública
- 2010-09-01: Aprovação do Conetrato de Cessão Onerosa e da Oferta Global
- 2010-08-27: Pagamento de Juros sobre Capital Próprio.
- 2010-07-16: Distribuição de Juros Sobre Capital Próprio.
- 2010-06-22: Capitalização e Cessão Onerosa
- 2010-06-21: Plano de Negocios 2010-2014
- 2010-06-21: Esclarecimento sobre notícias
- 2010-06-21: Petrobras e São Martinho anunciam parceria estratégica para a produção de etanol
- 2010-06-10: Fato Relevante - Aprovação da Cessão Onerosa e Capitalização da Petrobras
- 2010-05-26: Fato Relevante - Pagamento de JCP - 31.05.2010
- 2010-05-25: Convocação de Assembléia Geral Extraordinária
- 2010-05-14: Fato Relevante - Distribuição de Juros sobre Capital Próprio
- 2010-04-30: subject omitted (possible person's name); see the document at the CVM
- 2010-04-23: Pagamento de Dividendos e JCP
- 2010-01-22: subject omitted (possible person's name); see the document at the CVM
- 2010-01-14: Reservas Provadas da Petrobras em 2009 - Resumo
Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

Data quality
53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentamadequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da PetróleoBrasileiro S.A. - Petrobras em 31 de dezembro de 2010, o desempenho de suas operações e osseus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeisadotadas no Brasil.Opinião sobre as demonstrações contábeis consolidadas7."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten

consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 22 held-out cases, a single measurement with the frozen reader (dd688da): precision 0.973, coverage 0.915, literal check on 146 of 146 excerpts, no natural-person name (docs/leitor_medida_f1_2.json).

Explanatory notes · Related parties
Note: 11 Partes relacionadas (pages 26 to 34)
- [subsidiaries] “As operações comerciais da Petrobras com suas subsidiárias, controladas e sociedades de propósito específico são efetuadas a preços e condições normais de mercado. As operações de mútuo são realizadas de acordo com as condições de mercado e legislação aplicável.” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “CONTROLADAS (*) 0” (p. 26, original excerpt in Portuguese)
- Excerpt omitted; see page 26 of the document
- [subsidiaries] “Gaspetro 1.080 289 340 811 0 0 0 2.520” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “PifCo 3.189 0 0 0 0 4 0 3.193” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Downstream 207 0 0 0 183 0 0 390” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Transpetro 281 128 0 0 0 0 0 409” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Brasoil 0 0 0 0 26.603 7 0 26.610” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “BOC 0 0 0 0 30 1 0 31” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Petrobras Comercializadora Energia Ltda 65 45 0 0 0 0 0 110” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Petrobras Biocombustível S.A. 66 0 103 0 0 0 0 169” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “Termoelétricas 119 11 14 0 224 0 0 368” (p. 26, original excerpt in Portuguese)
- [subsidiaries] “7.490 1.461 465 811 28.085 77 275 38.664” (p. 26, original excerpt in Portuguese)
- [associates] “COLIGADAS 232 62 3 0 0 0 0 297” (p. 26, original excerpt in Portuguese)
- [associates] “31/12/2010 8.716 1.523 468 811 28.085 77 447 40.127” (p. 26, original excerpt in Portuguese)
- [associates] “31/12/2009 7.790 780 295 973 47.837 78 1.511 59.264” (p. 26, original excerpt in Portuguese)
- [joint ventures; subsidiaries] “(*) Inclui suas controladas e grupo de controladas em conjunto” (p. 26, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora)” (p. 27, original excerpt in Portuguese)
- [subsidiaries] “Brasil S.A. (TBG), controlada da Gaspetro. Em 31 de dezembro de 2010, o total de créditos da” (p. 27, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “CONTROLADAS (*)” (p. 28, original excerpt in Portuguese)
- Excerpt omitted; see page 28 of the document
- [subsidiaries] “Gaspetro (659) (318) -- - - - - (977)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “PifCo (9.514) (102) -- - - - (9.940)(324)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “PNBV (88) - (1.562) -- - - - (1.650)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “Downstream (145) -- - - - - - (145)” (p. 28, original excerpt in Portuguese)

- [subsidiaries] “Transpetro (561) -- - - - - - (561)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “Brasoil (158) - (7) -- - - - (165)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “Termoelétricas (204) -- (29) - (580) -- (813)” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “Marlim Participações S.A. -- - - - - - - -” (p. 28, original excerpt in Portuguese)
- [subsidiaries] “(430)(12.145) (1.569) (2.241) - (350)(2.836) (19.571)” (p. 28, original excerpt in Portuguese)
- [associates] “COLIGADAS (84) (2) -- - - (54) -(140)” (p. 28, original excerpt in Portuguese)
- [associates] “31/12/2010 (12.229) (432) (1.569) (3.151) (139) (14.924) (54) (350) (32.848)” (p. 28, original excerpt in Portuguese)
- [associates] “31/12/2009 (29.723) (752) (1.394) (3.502) (139) (10.904) (49) (856) (47.319)” (p. 28, original excerpt in Portuguese)
- [joint ventures; subsidiaries] “(*) Inclui suas controladas e grupo de controladas em conjunto” (p. 28, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 29, original excerpt in Portuguese)
- Excerpt omitted; see page 29 of the document
- [subsidiaries] “(288)88.207 (903) 87.016” (p. 29, original excerpt in Portuguese)
- [associates] “COLIGADAS 1411.304 (27) 11.291” (p. 29, original excerpt in Portuguese)
- [associates] “31/12/2010 99.641 (1.310) (778) 97.553” (p. 29, original excerpt in Portuguese)
- [associates] “31/12/2009 85.900 1.514 (6.603) 80.811” (p. 29, original excerpt in Portuguese)
- [joint ventures; subsidiaries] “(*) Inclui suas controladas e grupo de controladas em conjunto” (p. 29, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 30, original excerpt in Portuguese)
- Excerpt omitted; see page 30 of the document
- [subsidiaries] “A Petrobras tem como procedimento conceder garantias às subsidiárias e controladas para algumas operações financeiras realizadas no exterior.” (p. 30, original excerpt in Portuguese)
- [subsidiaries] “As garantias oferecidas pela Petrobras são efetuadas com base em cláusulas contratuais que suportam as operações financeiras entre as subsidiárias e terceiros, garantindo a compra da dívida em caso de inadimplência por parte das subsidiárias e controladas.” (p. 30, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 31, original excerpt in Portuguese)
- [subsidiaries] “Em 31 de dezembro de 2010, as operações financeiras realizadas por estas subsidiárias e” (p. 31, original excerpt in Portuguese)
- [subsidiaries] “11.6 Fundo de investimento no exterior de subsidiárias” (p. 31, original excerpt in Portuguese)
- Excerpt omitted; see page 31 of the document
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 32, original excerpt in Portuguese)
- [associates] “11.7 Transações com coligadas, entidades governamentais e fundos de pensão” (p. 32, original excerpt in Portuguese)
- Excerpt omitted; see page 32 of the document
- [associates; other related parties] “As transações significativas com coligadas, entidades governamentais e fundo de pensão resultaram nos seguintes saldos:” (p. 32, original excerpt in Portuguese)
- [associates] “Coligadas 305 144 950 166” (p. 32, original excerpt in Portuguese)
- [associates] “Braskem 84 60 594 76” (p. 32, original excerpt in Portuguese)
- [associates] “Quattor 78 43 260 41” (p. 32, original excerpt in Portuguese)

- [associates] “Outras Empresas Coligadas 143 41 96 49” (p. 32, original excerpt in Portuguese)
- [associates] “Entidades governamentais e fundos de pensão 42.824 56.007 16.650 49.156” (p. 32, original excerpt in Portuguese)
- [associates] “Banco do Brasil S.A. 5.067 9.415 1.484 7.294” (p. 32, original excerpt in Portuguese)
- [associates] “BNDES 3 36.320 1 34.929” (p. 32, original excerpt in Portuguese)
- [other related parties] “Petros (Fundo de Pensão) - 501 - 523” (p. 32, original excerpt in Portuguese)
- [other related parties] “Outros 221 423 (937) 509” (p. 32, original excerpt in Portuguese)
- [other related parties] “56.15143.129 17.600 49.322” (p. 32, original excerpt in Portuguese)
- [other related parties] “Circulante 34.481 8.393 10.394 5.982” (p. 32, original excerpt in Portuguese)
- [other related parties] “Não circulante 8.648 47.758 7.206 43.340” (p. 32, original excerpt in Portuguese)
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 33, original excerpt in Portuguese)
- Excerpt omitted; see page 33 of the document
- [parent company] “Notas explicativas às demonstrações contábeis (Consolidadas e da Controladora) (Em milhões de reais, exceto quando indicado em contrário)” (p. 34, original excerpt in Portuguese)
- [key management personnel] “valor de R$ 6.442.020,83 (R$ 5.248.780,31 em 2009) referente a sete diretores e nove” (p. 34, original excerpt in Portuguese)
- [key management personnel] “conselheiros.” (p. 34, original excerpt in Portuguese)
- [key management personnel] “No consolidado, os honorários da diretoria e do conselho de administração totalizam” (p. 34, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)

- Excerpt omitted; see page 26 of the document
- Excerpt omitted; see page 27 of the document
- [terms and conditions] “Indexador 31.12.2010 31.12.2009” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “TJLP + 5% a.a. 26 49” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “LIBOR + 1 a 3% a.a. 24.174 44.798” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “1,70% a.a. 183 224” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “101% do CDI 115 171” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “14,5% a.a. 78 77” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “IGPM + 6% a.a. 146 146” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “Outras Taxas 3.363 2.372” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “28.085 47.837” (p. 27, original excerpt in Portuguese)
- [balances] “Em 31 de dezembro de 2010, o saldo dos direitos ao fornecimento futuro, por conta do custo” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “incorrido na obra, até aquela data, acrescidos de juros de 10,7% a.a., é de R$ 252 (R$ 339 em 31” (p. 27, original excerpt in Portuguese)
- [balances] “de dezembro de 2009), sendo R$ 149 classificados no ativo realizável a longo prazo como” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “transportar 6 milhões de metros cúbicos de gás pelo prazo de 40 anos (TCO - Transportation” (p. 27, original excerpt in Portuguese)
- [terms and conditions] “de metros cúbicos de gás, pelo prazo de 40 anos (TCO), era de R$ 811 (R$ 973 em 31 de” (p. 27, original excerpt in Portuguese)
- [balances] “dezembro de 2009), e está classificado no ativo realizável a longo prazo, como contas a receber líquidas.” (p. 27, original excerpt in Portuguese)
- Excerpt omitted; see page 28 of the document
- Excerpt omitted; see page 29 of the document
- [terms and conditions] “Os valores investidos em títulos públicos do FIDC-NP estão registrados em caixa e equivalentes de caixa e títulos e valores mobiliários, em função dos seus

respectivos prazos de realização.” (p. 30, original excerpt in Portuguese)
- [amount of the transactions; terms and conditions] “Os encargos financeiros a apropriar sobre as operações de venda de direitos creditórios performados e/ou não performados estão registrados como outros ativos circulantes.” (p. 30, original excerpt in Portuguese)
- [amount of the transactions; balances] “As cessões de direitos creditórios performados estão classificadas como outros ativos circulantes, enquanto não compensados. As cessões de direitos creditórios não performados estão registradas como outras contas e despesas a pagar no passivo circulante.” (p. 30, original excerpt in Portuguese)
- [terms and conditions] “Aplicações financeiras 3.442206 Títulos e valores mobiliários 1.7187.758 Encargos financeiros a apropriar 356426” (p. 30, original excerpt in Portuguese)
- [balances] “Total classificado no ativo circulante 7.768 4.678” (p. 30, original excerpt in Portuguese)
- [balances] “Cessões de direitos não performados (15.933) (14.318)” (p. 30, original excerpt in Portuguese)
- [balances] “Total classificado no passivo circulante (15.933) (14.318)” (p. 30, original excerpt in Portuguese)
- [amount of the transactions] “Receita financeira 184 1.397” (p. 30, original excerpt in Portuguese)
- [guarantees] “11.5 Garantias obtidas e concedidas” (p. 30, original excerpt in Portuguese)
- [guarantees] “A Petrobras tem como procedimento conceder garantias às subsidiárias e controladas para algumas operações financeiras realizadas no exterior.” (p. 30, original excerpt in Portuguese)
- [amount of the transactions; guarantees] “As garantias oferecidas pela Petrobras são efetuadas com base em cláusulas contratuais que suportam as operações financeiras entre as subsidiárias e terceiros, garantindo a compra da dívida em caso de inadimplência por parte das subsidiárias e controladas.” (p. 30, original excerpt in Portuguese)
- [balances; guarantees] “garantidas pela Petrobras apresentam os seguintes saldos a liquidar:” (p. 31, original excerpt in Portuguese)
- Excerpt omitted; see page 31 of the document
- [amount of the transactions; terms and conditions] “Gás S/A - Gaspetro e a remuneração cobrada está fixada em 0,30% ao ano, sobre o montante dos” (p. 31, original excerpt in Portuguese)
- [balances] “empresas que são consolidadas, foram compensados no saldo de financiamentos nos passivos circulante e não circulante.” (p. 31, original excerpt in Portuguese)
- [balances] “As transações significativas com coligadas, entidades governamentais e fundo de pensão resultaram nos seguintes saldos:” (p. 32, original excerpt in Portuguese)
- Excerpt omitted; see page 33 of the document
- [balances] “Ativo Circulante 34.481 10.394” (p. 33, original excerpt in Portuguese)
- [balances] “Caixa e equivalentes de caixa 5.424 8.369” (p. 33, original excerpt in Portuguese)
- [balances] “Títulos e valores mobiliários 25.525 -” (p. 33, original excerpt in Portuguese)
- [balances] “Contas a Receber, líquidas 3.392 1.784” (p. 33, original excerpt in Portuguese)
- [balances] “Outros ativos circulantes 140 241” (p. 33, original excerpt in Portuguese)
- [balances] “Não Circulante 8.648 7.206” (p. 33, original excerpt in Portuguese)
- [balances] “Conta petróleo e álcool - STN 822 817” (p. 33, original excerpt in Portuguese)
- [balances] “Títulos e valores mobiliários 5.177 4.583” (p. 33, original excerpt in Portuguese)
- [balances] “Depósitos judiciais 2.468 1.716” (p. 33, original excerpt in Portuguese)
- [balances] “Outros ativos realizados a longo prazo 181 90” (p. 33, original excerpt in Portuguese)
- [balances] “Passivo Circulante 8.393 5.982” (p. 33, original excerpt in Portuguese)
- [balances] “Financiamentos 3.667 2.836” (p. 33, original excerpt in Portuguese)
- [balances] “Dividendos propostos 1.596 691” (p. 33, original excerpt in Portuguese)
- [balances] “Outros passivos circulantes 3.130 2.455” (p. 33, original excerpt in Portuguese)
- [balances] “Passivo Não Circulante 47.758 43.340” (p. 33, original excerpt in Portuguese)
- [balances] “Financiamentos 47.634 43.210” (p. 33, original excerpt in Portuguese)
- [balances] “Outros passivos não circulantes 124 130” (p. 33, original excerpt in Portuguese)
- [balances] “56.15143.129 17.600 49.322” (p. 33, original excerpt in Portuguese)
- [balances] “O saldo desses recebíveis em 31 de dezembro de 2010 era R$ 3.145 (R$ 2.007 em 31

de dezembro” (p. 33, original excerpt in Portuguese)
- [balances] “de 2009), apresentados no ativo não circulante, e classificados como recebíveis de partes relacionadas, dos quais R$ 2.372 estavam vencidos.” (p. 33, original excerpt in Portuguese)
- [amount of the transactions] “A Companhia tem feito cobranças sistemáticas aos devedores e a própria Eletrobrás e pagamentos parciais têm sido realizados.” (p. 33, original excerpt in Portuguese)
- [amount of the transactions] “No consolidado, os honorários da diretoria e do conselho de administração totalizam” (p. 34, original excerpt in Portuguese)

Market terms
- “As operações comerciais da Petrobras com suas subsidiárias, controladas e sociedades de propósito específico são efetuadas a preços e condições normais de mercado. As operações de mútuo são realizadas de acordo com as condições de mercado e legislação aplicável.” (p. 26, original excerpt in Portuguese)

The sentence about market terms is recorded as a fact, without proof (CPC 05 (R1), item 23).

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- [emphasis of matter] “Ênfase 8. Conforme descrito na Nota Explicativa nº 2, as demonstrações contábeis individuais foram elaboradas de acordo com as práticas contábeis adotadas no Brasil. No caso da Petróleo Brasileiro S.A. - Petrobras essas práticas diferem do IFRS, aplicável às demonstrações contábeis separadas, somente no que se refere à avaliação dos investimentos em controladas, coligadas e controladas em conjunto pelo método de equivalência patrimonial, enquanto que para fins de IFRS seria custo ou valor justo; e pela opção pela manutenção do saldo de ativo diferido, existente em 31 de dezembro de 2008, que vem sendo amortizado.” (original excerpt in Portuguese)
- Excerpt omitted; see the auditor's report in the document

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, a header line (years, column titles) may still come out as an excerpt: in the reader's measurement, 1 of 19 header texts of the held-out cases came out as excerpts.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why

the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-2 (commit dd688da). The generation date is in the demo manifest (demo/manifesto.json).

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: PETRÓLEO BRASILEIRO S.A. - PETROBRAS.